Erratum (19 October 2026)

This page was added in front of the original report, which follows unchanged. It corrects one claim that appears in Section 1 (“37 corners can be built”), Section 3 (“Can one actually be built?”), Section 6 and the caption of Table 9 (“the 37-corner one is practical”).

The 26- and 37-corner bodies modelled here cannot be built as described. The model has a 0.5 mm polycarbonate shell and a 5.2 g tungsten weight in each corner, “its centre set 3 mm inside the corner”. But the weight is treated as a point. A real 5.2 g tungsten piece is an 8.0 mm ball or a 6.5 mm cube, and at 9 cm it does not fit there:

- a weight centre 3 mm inside the corner is only 1.1–2.5 mm (p26) or 1.5–2.5 mm (p37) from the face planes, so the weight would stick out through the shell;
- near the base, neighbouring corners are only 4.4–4.6 mm (p26) and 2.8–3.8 mm (p37) apart, so the weights would overlap (15 and 24 pairs of weight centres closer than 8 mm).

The masses (142 g, 201 g) and the margins on the fit's own measure (1.43 mm, 3.36 mm) are reproduced. Measured as the distance the centre of mass must move, the nearest corner margin is 0.71 mm (p26) and 2.41 mm (p37). The tolerance figures (Table 4) are correct for the idealised model, not for a real build. The statements that “a careful maker could plausibly build the 37-corner body” and that “a 37-corner shell is” buildable are withdrawn for this design. The mass-distribution scale of Section 7 (Table 9) is unaffected as mathematics, but it says nothing about whether real weights fit.

A buildable weighted 37-corner design now exists, but only on paper (designed and simulated, not yet printed). It has 37 tungsten cylinders (3/8″ × 0.44″, 0.5 oz) in real pockets in a 0.8 mm PLA print, 200 mm tall, 626 g, with the rings refitted for that mass distribution. Its smallest margin is 4.16 mm. With face-plane errors of 0.1 / 0.2 / 0.3 / 0.5 mm, 100 / 95 / 81 / 27% of simulated builds keep one stable face. Files, build guide and scripts: exhibits/Bille/print/w37/ and exhibits/Bille/reconstruction/w37/ (fitcheck.py holds the check above).

In the Bille Drop exhibit the 26- and 37-corner bodies are still simulated with the idealised model's mass properties, and are now labelled as idealised designs.

One resting point is not enough

Gömböc-like bodies built and tested in Bullet Physics Playground: the spiral polyhedra of Domokos & Kovács, Sloan's analytic Gömböcs, and Gomboc-C

Working notes for W. M. Spears, prepared with Claude (Anthropic) · 9 October 2026 (revised) · code: demo/WyomingWill/Gomboc and /Bille in github.com/bullet-physics-playground/bpp

Summary

- **The 21-vertex polyhedron is the fewest corners, not the best design.** In the Conway (n,k)-spiral family (an apex above n horizontal regular k-gons, equal masses at the corners) the tipping margins grow quickly with more corners: about 0.8 mm (21 corners), 2.8 mm (26) and 4.3 mm (37) for a body 9 cm tall. Our fits for 13, 16, 17 and 19 corners all failed, consistent with the paper's conjecture that 21 is the minimum for this family.
- **The 21-corner body can only be an idealisation; 37 corners can be built.** Modelling a real object (a thin polycarbonate shell with a tungsten weight in each corner), the 21-corner body fails unless the skin is about 0.1 mm thick and the weights sit right in the corners. A 37-corner body refitted for a 0.5 mm shell and 5.2 g weights centred 3 mm inside the corners keeps a 3.4 mm margin and still works in 98% of trials with 0.5 mm build errors.
- **We could not reproduce the paper's centre-of-mass values.** Building the spirals literally from the angles in Table 1 of Domokos & Kovács, none of the five rows gives the listed z_C, and neither reading of the 3D construction we tried is mono-monostatic. This is most likely our misreading of the construction, but it is worth asking the authors.
- **Sloan's analytic Gömböcs are mathematically monostatic but poor self-righters.** At visible lumpiness the surface is not convex and the convex hull (what touches the table) gains a second resting basin above $\beta \approx 0.09$ (form 1) and $\beta \approx 0.05$ (form 2). Below that, the height difference between resting and balancing points is only 3.5–5 mm at 9 cm (Gomboc-C: 11 mm), and nearly level shelves 20–45° from home stall them: at realistic rolling resistance only 4–9% come to rest upright, and no setting makes them reliable (Tables 5a–5b).
- **Bille and the spiral polyhedra are one class: bodies that work because of where their mass is.** In the spiral family, the design works only with the mass in the corners; we found no wire-frame, hollow-shell or uniform-solid version up to 65–101 corners. A corner-weighted design tolerates up to 1.8% (21 corners), 9.7% (26) or 17.9% (37) of its mass spread over the faces. Section 8 lists eleven research topics, three of which follow directly from these results.
- **Simulation check:** in 117 drops of the three polyhedra, 116 ended on their stable face (21 ideal: 39/39; 26 shell: 37/38; 37 shell: 40/40).

1 The 21-vertex polyhedron and Table 1

Domokos & Kovács (2023) build a mono-monostatic 0-skeleton (equal point masses at the vertices, nothing else) from a Conway n-spiral: points $P_0 \ldots P_n$ in a vertical plane, $P_0 = (0,0,1)$, each radius OP_i perpendicular to the edge $P_{i-1}P_i$, with central angles α_i. The spiral is rotated k times about the vertical axis and the copies of each P_i form a horizontal regular k-gon, giving $nk + 1$ vertices. The paper gives the angles and z_C (Table 1) but no vertex coordinates.

We rebuilt each row in two ways: with the polygon vertices on the spiral, and with the spiral passing through the polygon edge midpoints (vertex radius x/cos(π/k)). Then we computed the centre of mass from the paper's formula and counted equilibria exactly (stable faces whose foot point lies inside the face; unstable vertices that are the unique farthest point in their direction).

(n, k)	v	z_C in Table 1	z_C from the angles (eq. 8)	stable / unstable: vertices on spiral	edge midpoints on spiral
(2, 25)	51	−0.000513	−0.000817	26 / 1	1 / 1
(3, 8)	25	−0.006141	−0.012412	17 / 1	1 / 9
(4, 5)	21	−0.015354	−0.036388	11 / 1	1 / 6
(5, 4)	21	−0.029972	−0.065455	13 / 1	1 / 5
(7, 3)	22	−0.042695	−0.106308	7 / 1	1 / 7

Table 1. Our literal reconstruction of Domokos & Kovács' Table 1. The ratio between the listed and computed z_C varies from row to row, so this is not a single typo; we have probably misread some detail of the construction.

Because the literal reading did not work, we kept the structure and re-fitted the 2n ring radii and heights to maximise the smallest margin: the distance by which the centre of mass's foot point misses every face but the stable one, and the amount by which every vertex but the apex fails to be the farthest point in its own direction. The resulting 21-vertex body stays within about 0.02 (relative) of the paper's spiral and has a 0.83 mm margin at 9 cm tall.

2 Why 21? The rest of the spiral family

The paper calls 21 an upper bound for the fewest vertices of a mono-monostatic 0-skeleton, and believes it is the minimum for rotationally symmetric spirals; its Table 1 also lists 22-, 25- and 51-vertex members. We fitted other (n,k) with the same margin-maximising search (20 random starts each, Nelder–Mead):

(n, k)	(3, 4)	(3, 5)	(4, 4)	(3, 6)	(4, 5)	(5, 4)	(5, 5)	(6, 6)
vertices	13	16	17	19	21	21	26	37
best margin (mm, 9 cm tall)	−0.06	−0.07	0.00	−0.06	0.83	0.62	2.78	4.27
	fails	fails	fails	fails	works	works	works	works

Table 2. Best margins found per (n,k). A failure here means our search found no working shape, not a proof that none exists.

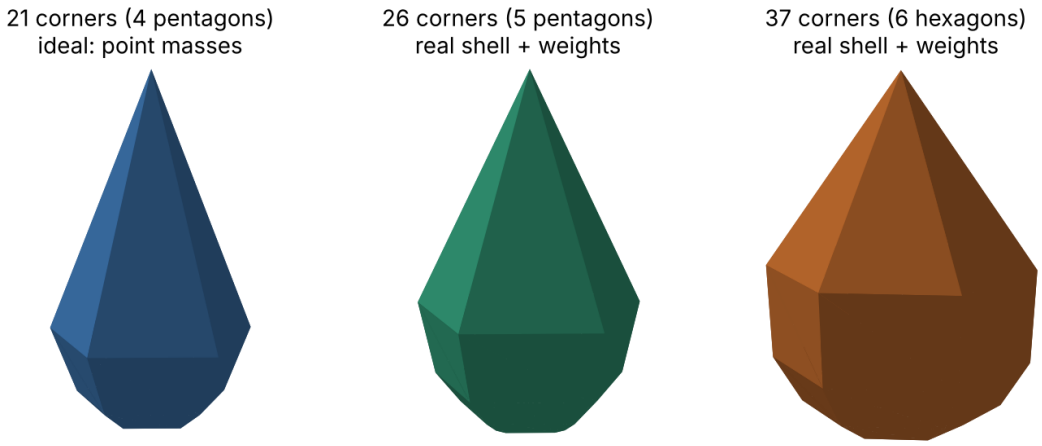


Figure 1. The three bodies used in the simulation, drawn as closed polyhedra, 9 cm tall. The 21-corner one is simulated as the paper's idealisation; the 26- and 37-corner ones as real shells with corner weights (Section 3).

3 Can one actually be built?

A physical version needs faces and weights that are not points. We modelled the simplest build: a polycarbonate shell (1.2 g/cm³) covering every face, with a 5.2 g tungsten weight (an 8 mm ball) in each corner, its centre set some distance inside the corner towards the body's middle. The shell pulls the centre of mass towards the faces' centroid and the inset pulls the weights inward; both eat into the margins.

build (9 cm tall, 5.2 g weights)	21 corners	26 corners	37 corners
point masses only	+0.83	+2.78	+4.27
0.1 mm skin, weights at the corners	+0.37	+2.57	+4.06
0.1 mm skin, weights 2 mm in	−0.95	+2.08	+3.54
0.3 mm skin, weights 2 mm in	−1.85	+1.45	+3.14
0.5 mm skin, weights 2 mm in	−2.72	+0.83	+2.75
0.5 mm skin, weights 4 mm in	−4.03	+0.13	+2.23
1.0 mm skin, weights 4 mm in	−6.05	−1.32	+1.31

Table 3. Margins in mm (positive = exactly one stable and one unstable equilibrium) for the point-mass shapes of Table 2, not refitted. Refitting the 21-corner shape for each build helps only a little: with a 0.3 mm skin and weights 2 mm in, it needs about 20 g per corner (420 g in all) to scrape by with a 0.05 mm margin.

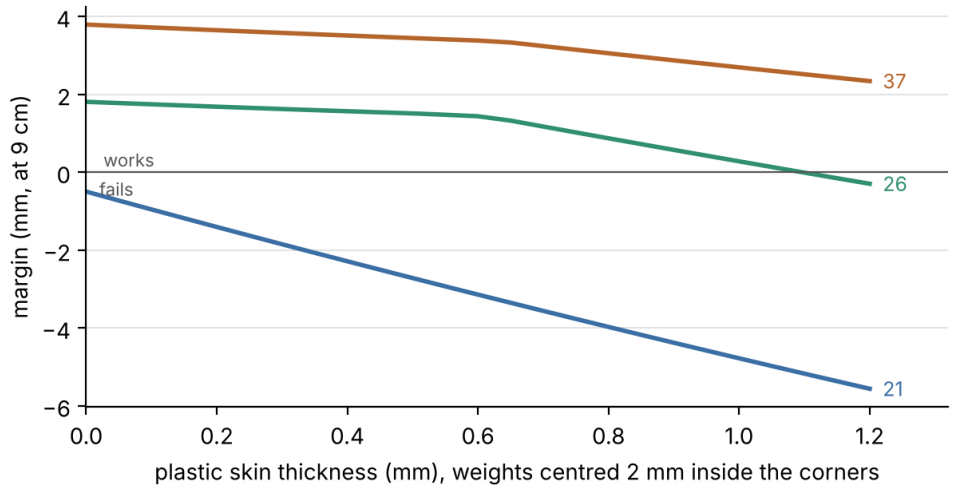


Figure 2. Margin against skin thickness with the weights 2 mm inside the corners, for the shapes actually used in the simulation (21: point-mass fit; 26 and 37: refitted for a 0.5 mm skin with weights 3 mm in).

Refitted for a real build (0.5 mm skin, 5.2 g weights centred 3 mm inside each corner) the 26-corner body keeps a 1.43 mm margin (142 g) and the 37-corner body a 3.36 mm margin (201 g). Margin is not the whole story, though: what matters in a workshop is tolerance to build errors. We assumed accurate flat panels, ring radii, ring heights and weight positions each off by a random amount, and 3% scatter in the weights, over 300 random trials per row:

errors in rings and weight positions	0.2 mm	0.5 mm	1.0 mm
21 corners, ideal (point masses)	78%	22%	2%
26 corners, shell + weights	75%	27%	5%
37 corners, shell + weights	100%	98%	67%

Table 4. Share of trials that still have exactly one stable face and one unstable vertex. Corner-by-corner errors that warp the flat panels are far less forgiving: small faces tilt, and their foot points move by roughly height × tilt.

So a careful maker could plausibly build the 37-corner body, the 26 would be demanding, and the 21 is essentially a mathematical object. That is our reading of these numbers; the paper does not discuss physical realisation.

4 Sloan’s analytic Gömböcs

Sloan (2023, via MathWorld) gives Gömböcs in closed form: $r^4 = 1 + 4\beta \sin\varphi \cos(\theta - 5\varphi)$ (form 1) and $r^4 = 1 + 4\beta \sin\varphi \cos(\theta - (3\pi/2)(\cos\varphi - \cos^3\varphi/3))$ (form 2), where φ is measured from the pole. Using r^4 puts the centroid exactly at the origin, and on each circle of latitude the radius peaks at a single θ, which makes it easy to show that r has exactly one maximum and one minimum. MathWorld lists form 2’s unstable point at $\varphi = 0$, but r is largest at $\varphi = \pi/2$, $\theta = 0$, opposite the resting point.

A 2026 reanalysis (arXiv:2604.17120) finds these surfaces strictly convex only for β below about 0.036, i.e. barely distinguishable from a ball. A body rolls on its convex hull, so we asked how far β can go before the *hull* gains a second resting basin. We counted basins by persistence of face heights on the hull’s face-adjacency graph, taking the barrier between neighbouring faces as the highest the centre of mass rises when rolling over their shared edge. Gomboc-C’s hull gives exactly one basin.

β	0.04	0.05	0.06	0.07	0.08	0.10	0.12
form 1: second basin depth (µm)	—	—	—	—	—	67	185
form 2: second basin depth (µm)	—	59	213	424	670	1210	1791
form 1: balance − rest height (mm)	3.5	4.0	4.5	4.8	5.1	5.6	6.2

Table 5. Second resting basins on the convex hull of Sloan's shapes at 9 cm ("—" in the first two rows: none deeper than the ~6 µm discretisation noise). Gomboc-C's balancing point is 11.3 mm above its resting height.

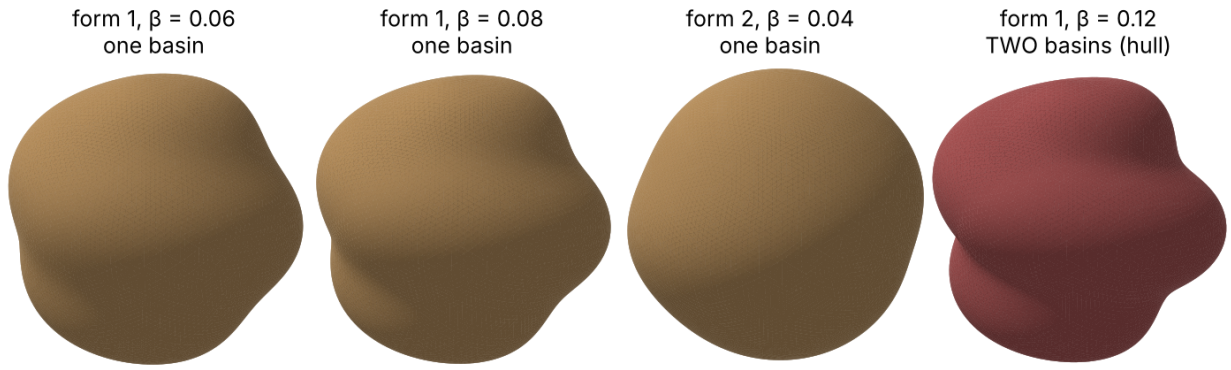


Figure 3. The lumpiest Sloan shapes whose hull keeps one resting basin (left three) and one that does not (right).

Why they rarely get home. For every orientation we measured gravity's righting lever: the horizontal distance between the contact point and the point below the centre of mass. Wherever it is smaller than the rolling-resistance lever arm, a body can simply stop. (The Sloan shapes' true resting orientation lies 4–5° from the formula's minimum, on a hull face bridging a dimple.)

tilt from home	Gomboc-C: weakest lever	Sloan β 0.07 / 0.08: weakest lever
5–10°	9.3 mm	10.1 / 10.7 mm
10–20°	7.8 mm	6.1 / 6.2 mm
20–30°	6.5 mm	1.5 / 1.8 mm
30–45°	0.07 mm, in small patches (0.2% of orientations)	0 mm, on broad shelves (0.6% of orientations)

Table 5a. The smallest righting lever found in each band of tilt. Gomboc-C pulls hard all the way from 30° to home; the Sloan shapes have nearly level shelves 20–45° from home, and their resting and balancing points differ in height by only ~5 mm (Gomboc-C: 11 mm).

A rolling-resistance sweep (six bodies at a time, each re-dropped as soon as it came to rest):

rolling resistance	Sloan β 0.07 / 0.08: runs finished	upright	median time to rest	Gomboc-C: runs finished	upright	median time to rest
0	none in 3 min	—	—	none in 3 min	—	—
10 µm	10 (in 20 min)	3	62 s	none in 4 min	—	—
20 µm	12	0	43 s	3	3	163 s
50 µm	47	2 (4%)	18 s	6	6	32 s
100 µm	120	7 (6%)	7.4 s	—		
150 µm	157	14 (9%)	5.4 s	117 (earlier tests)	all	10–20 s

Table 5b. Runs of 4 simulated minutes per setting (10 µm: 20 minutes in all). Without rolling resistance nothing ever settles, Gomboc-C included: nothing removes the energy of the drop.

There is no good middle. Rolling resistance small enough not to strand a Sloan shape on a shelf drains its energy so slowly that it rolls for minutes; anything larger stops it quickly, usually on a shelf 60–120° from home. Gomboc-C has no such dilemma: whatever setting drains its energy also brings it home, every time. One resting point in theory is not enough; the slopes leading to it decide whether a body gets there. That is what the real Gömböc's steep, sharp-edged design buys.

5 Simulation results

body	mass	runs	upright	median time to rest
Gomboc-C at 75% / 100% / 125%	136 / 323 / 630 g	every test	all	upright from a headstand in 8.8 / 13.2 / 14.3 s
21 corners, ideal	109 g	39	39	4.3 s
26 corners, shell + weights	142 g	38	37 (one stopped 21° off)	2.3 s
37 corners, shell + weights	201 g	40	40	2.5 s
Sloan β 0.07 / 0.08	407 / 401 g	157	14 (9%)	5.4 s

Table 6. Random drops on the wooden tray at 1200 physics steps per second, convex-hull contact, rolling resistance 150 µm and spin friction 0.7 rad/s². The one 26-corner failure was not touching anything; we have not investigated it.

The demos are now split by this classification. **Gomboc Variety** (Gomboc/gomboc-variety.lua) holds the uniform-density bodies: Gomboc-C at three sizes and Sloan β 0.07 and 0.08. **Bille Drop** (Bille/bille-drop.lua) holds the weighted ones: Bille and the 21-, 26- and 37-corner polyhedra, scaled to 40 cm tall to stand beside the 50 cm Bille, with each body's path of faces printed (e.g. S3 → S2 → S1 → base); half the colours are always Billes, the rest cycle through the polyhedra, and a Bille's edge lights up while it lies flat on the table. In Gomboc Variety a white dot marks each body's balancing point; at home it is at or near the top (within ~1° on Gomboc-C, ~5° on β 0.07, ~35° on β 0.08, whose one resting valley is long and nearly level). In 40 drops in the Bille demo every body that came to rest untouched ended on its stable face.

6 Is any of this new, and is it publishable?

Honestly: some of it may be new, none of it is a major mathematical result, and all of it needs checking before anyone claims it.

- **Probably not new:** that larger spirals work (the paper's Table 1 already lists 22, 25 and 51 vertices) and that 21 is the minimum for this family (the paper's own belief; our failed fits are weak supporting evidence, not proof).
- **Possibly new, and the most useful:** the margin and build-tolerance analysis. The margins grow from 0.8 to 4.3 mm between 21 and 37 corners, the 21-corner body is not physically buildable while a 37-corner shell is, and the tolerance figures of Table 4. We found no discussion of physical realisation in the paper or in the sources we searched; that search was brief.
- **Worth raising with the authors:** that the Table 1 centre-of-mass values could not be reproduced from the stated construction. Most likely we misread it, but the answer would help anyone trying to build these.
- **Small but concrete:** the hull-based limits on β for Sloan's shapes (Table 5), complementing the convexity limit in arXiv:2604.17120, and the correction to MathWorld's form-2 unstable point.

A suitable outlet would be a short note, such as an arXiv preprint or a piece for *The Mathematical Intelligencer*, *Math Horizons* or a maker-oriented venue, ideally paired with a 3D-printed 37-corner model. The best first step is to write to Gábor Domokos and Flórián Kovács (BME Morphodynamics group): they will know at once whether the build-tolerance question has been answered, and whether we misread their construction.

Before submitting anything: (1) replace the floating-point margins with certified (interval or exact rational) equilibrium counts; (2) settle the Table 1 reading; (3) repeat the tolerance study with realistic panel errors and a real weight geometry; (4) do a proper literature search, including the Conway–Guy and Bezdek lines of work on monostable polyhedra.

7 One class: it is all about where the mass goes

Gomboc-C is a uniform solid. Bille and the spiral polyhedra are not: they work because of where their mass is put. The literature already sorts monostatic polyhedra both by their numbers of stable and unstable equilibria (S, U) and by how mass may be distributed. Read that way, Bille and the 21-corner body are two points on one scale, and the less freedom there is in placing the mass, the more faces and vertices are needed:

where the mass may go	best known results	source
anywhere inside (free centre of mass)	Bille: 4 vertices, class (1, 2) — one stable face, two unstable vertices (our reconstruction confirms 1 / 2). In 20,000 random tetrahedra with random centres of mass, none was (1, 1).	Almadi, Dawson & Domokos 2025; this work
equal masses at the vertices (0-skeleton)	(1, 1): 21 vertices known, at least 8 needed. Mono-unstable: 11 vertices and 8 faces is the minimum.	Domokos & Kovács 2023; Bozóki et al. 2021, 2023
uniform along the edges (1-skeleton, wire frame)	nothing found	—
uniform over the faces (2-skeleton, hollow shell)	nothing found	—
uniform solid (3-skeleton)	monostable: 14 faces known (Reshetov), 18 vertices; mono-unstable: 18 faces and vertices known, at least 7 vertices; (1, 1): exists (Lángi 2022), no example known. Smooth: the Gömböc.	Conway & Guy 1969; Bezdek 2011; Reshetov 2014; Bozóki et al. 2024; Regős 2025

Table 7. The mass-distribution scale. “Nothing found” means we found no result in a brief search, not that none exists.

Our searches. We fitted the Conway (n,k)-spiral family under each mass model, maximising the margin as before (12 random starts per case):

mass model	corners tried	result
vertices (point masses)	13, 16, 17, 19 / 21, 26, 37	fails / works (0.8, 2.8, 4.3 mm)
free centre of mass, kept inside	4, 5, 7, 9, 11 / 10, 13	fails / works, but only with the centre of mass about 0.1 mm above the bottom face (a “weighted base”)
edges (wire frame)	21, 25, 26, 31, 37, 50, 65	fails every time (5–41 extra equilibria)
faces (hollow shell)	21, 25, 26, 31, 37, 50, 65	fails every time
uniform solid	37, 65, 101	fails every time

Table 8. Best fits in the spiral family. The spiral works only when the mass sits at the corners: spreading it over edges or faces raises the centre of mass too far. Wire-frame and shell versions would need a different construction, if they exist at all.

How much evenly spread mass can a corner-weighted design carry? Mixing the two models, with a share f of the mass spread evenly and the rest in equal corner weights:

design (point-mass fit)	spread over the faces	spread along the edges
21 corners	up to 1.8%	up to 2.1%
26 corners	up to 9.7%	up to 13.3%
37 corners	up to 17.9%	up to 24.7%

Table 9. Largest evenly spread share of the mass that keeps exactly one stable and one unstable equilibrium. This one number explains why the 21-corner body is unbuildable and the 37-corner one is practical, and it gives a single scale running from the point-mass ideal towards uniform bodies.

8 Research topics we might contribute to

Ranked roughly by how well they fit what we already have: a physics simulator, shape optimisation, exact equilibrium counting, and the means to 3D-print and weigh models. “Uniform” means uniform density, the true Gömböc problem; “weighted” means Bille-type bodies.

#	topic	density	what we would do	fit
1	**The mass-distribution scale.** Fewest vertices for (1, 1) when the mass is on the edges or the faces; how much evenly spread mass a design can carry (Tables 8–9).	weighted → uniform	new constructions beyond the spiral; trade-off curves; certified counts	strong
2	**Buildable discrete Gömböcs.** Designs that survive real shells, weights and tolerances; printed and weighed prototypes compared with simulation.	weighted	37-corner prototype, tolerance study with realistic panel errors	strong
3	**Self-righting dynamics.** What makes a monostatic body actually get home against rolling resistance: stall regions, righting time against size and shape (Sloan shapes stall, Gomboc-C does not); links to turtle shells.	both	simulation sweeps; a “righting quality” measure; check against filmed real Gömböcs	strong
4	**Sloan-type analytic shapes.** How far analytic families can go before the convex hull gains extra resting points, and families designed for steep, fast righting.	uniform	hull basin counts (Table 5), new analytic families	good
5	**Monostable tetrahedra (Bille family).** Is (1, 1) impossible for any tetrahedron with any mass distribution? How big can the loading zones be?	weighted	proof or certified search; zone optimisation	good
6	**Fewest vertices with a free centre of mass.** Smallest (1, 1) polyhedron when the mass may be placed anywhere, excluding the trivial weighted base.	weighted	exhaustive search with exact counting	medium
7	**Certified tools and an open dataset.** Interval/exact equilibrium counting and coordinates for every known monostatic polyhedron.	both	software release; reproduce published constructions, incl. Table 1	good
8	**Simulation methods for near-degenerate contact.** Energy injection, time steps and rolling-resistance models in rigid-body engines, with the Gömböc as a benchmark.	both	the Bullet studies we have already done, written up	good
9	**An explicit uniform (1, 1) polyhedron.** Lángi proved one exists; none is known.	uniform	large optimisation searches	long shot
10	**Lower bounds for uniform monostatic polyhedra** (7 vertices for mono-unstable is the current bound).	uniform	needs semidefinite certification expertise	long shot
11	**Exhibits and teaching.** The Euler’s Cabinet gallery and these simulations as interactive explanations.	both	already under way	strong

Table 10. Possible contributions. Topics 1–3 build directly on results in this report and look the most promising; 9–10 are major open problems that specialists are already working on.

Before investing in any of these, the right first step is still the one in Section 6: a short note to the Domokos group setting out Tables 4, 8 and 9 and the Table 1 question, asking what is already known.

9 Methods and limitations

Equilibria of polyhedra were counted exactly in double precision; margins are distances in cm at 9 cm height. Hull basins for the smooth shapes used a 25,000-point surface sample; mass properties used a 164,000-vertex mesh. Simulation: Bullet 3.24 in Bullet Physics Playground, with script-side rolling resistance and spin friction applied before every step. All scripts are in demo/WyomingWill/Gomboc/variety-reconstruction. Limitations: no certified arithmetic; the shell model treats faces as thin plates and weights as spheres; the robustness model assumes perfectly flat panels; the β-shape behaviour depends on the rolling-resistance model.

References

G. Domokos, F. Kovács, “Conway’s spiral and a discrete Gömböc with 21 point masses”, *Amer. Math. Monthly* 130 (2023); arXiv:2103.13727.

M. L. Sloan, “An Analytical Gomboc”, arXiv:2306.14914 (2023); summarised on MathWorld, “Gömböc”.

“Sloan’s Analytical Gömböc at Published β: A Strict-Convexity-Constrained Reanalysis”, arXiv:2604.17120 (2026).

P. L. Várkonyi, G. Domokos, “Mono-monostatic bodies: the answer to Arnold’s question”, *Math. Intelligencer* 28 (2006).

Z. Lángi, “A solution to some problems of Conway and Guy on monostable polyhedra”, *Bull. London Math. Soc.* 54 (2022).

S. Bozóki, G. Domokos, F. Kovács, K. Regős, “Mono-monostatic polyhedra with uniform point masses have at least 8 vertices”, *Int. J. Solids Struct.* (2021); arXiv:2103.12769.

S. Bozóki, G. Domokos, F. Kovács, K. Regős, “The smallest mono-unstable convex polyhedron with point masses has 8 faces and 11 vertices”, *Eur. J. Oper. Res.* 310 (2023); arXiv:2302.04252.

S. Bozóki, G. Domokos, D. Papp, K. Regős, “The smallest mono-unstable, homogeneous convex polyhedron has at least 7 vertices”, arXiv:2401.17906 (2024).

K. Regős, “Homogeneous, monostatic lattice polyhedra, or the magic pyramid with 17 sides”, *Építés – Építészettudomány* 53 (2025).

G. Almadi, R. J. MacG. Dawson, G. Domokos, “Building a monostable tetrahedron”, arXiv:2506.19244 (2025).

G. Domokos, P. L. Várkonyi, “Geometry and self-righting of turtles”, *Proc. R. Soc. B* 275 (2008).